A Simple Randomization Test for the Presence of Interference

Ye Wang[1] Weihan Zhang[2] Peng Ding[3]

[1]Political Science, UNC Chapel Hill [2]Biostatistics, UC Berkeley [3]Statistics, UC Berkeley

Motivation and contribution

- **Motivation.** Conditional randomization tests (Athey et al., 2018; Puelz et al., 2022) for interference are finite-sample exact under partially sharp nulls but require carefully chosen focal units.
- Their selection is cumbersome, and using only this subset can reduce power.
- **Contribution.** We develop studentized (Wu & Ding, 2021), full-sample randomization tests without focal selection.
- Regression enables joint testing across proximity levels, while Gaussian prepivoting (Cohen & Fogarty, 2022) targets localized interference.
- We derive sampling and randomization variances to establish asymptotic validity, sacrificing finite-sample exactness.

Testing interference beyond $\tilde{d}$

Allow direct effects and arbitrary effects within $\tilde{d}$:

$$H_0(\tilde{d}): \quad Y_i(z) = Y_i(\tilde{z})$$
$$\text{whenever } z_{\mathcal{N}_i(\tilde{d})} = \tilde{z}_{\mathcal{N}_i(\tilde{d})}.$$

Implied weak nulls. Average spillover effects vanish beyond $\tilde{d}$:

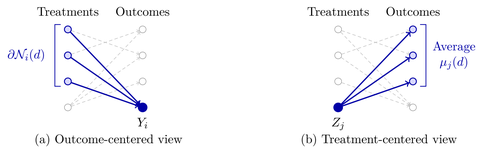
$$H_0(\tilde{d}) \Longrightarrow H_0^w(d): \ \tau(d) = 0, \qquad d > \tilde{d}.$$

A full-sample HT statistic

Independent $Z_j \sim \text{Bern}(p_j)$; test $d > \tilde{d}$.

$$\tilde{Z}_j = \frac{Z_j - p_j}{p_j(1-p_j)}, \qquad \mu_j(d) = \frac{\sum_{i:d_{ij}=d} Y_i}{|\partial\mathcal{N}_j(d)|}.$$

$$T(d) = \frac{1}{N}\sum_j \tilde{Z}_j \mu_j(d), \qquad \mathbb{E}[T(d)] = \tau(d).$$



(a) Outcome-centered view (b) Treatment-centered view

The dual view preserves the same neighborhood weights. Empty-neighborhood averages are zero. HT construction: Wang et al. (2025).

Sampling and randomization variances

Under $H_0(\tilde{d})$ and $\tilde{d} < d \le \bar{d}$,

$$\text{Var}[T(d)] = \frac{1}{N^2}\sum_j \frac{\mathbb{E}[\mu_j^2(d)]}{p_j(1-p_j)} + \frac{1}{N^2}\sum_{j\neq l} \mathbf{1}\{\mathcal{N}_j(\bar{d}+\tilde{d}) \cap \mathcal{N}_l(\bar{d}+\tilde{d}) \neq \varnothing\} \times \mathbb{E}[\tilde{Z}_j\tilde{Z}_l\mu_j(d)\mu_l(d)].$$

- The HAC-style estimator retains all overlapping pairs.
- **Artificial sharp null.** Under H_0^*, impute $Y_i(z) = Y_i$ for all i, z (Wu & Ding, 2021). Holding outcomes fixed and redrawing treatments gives

$$\text{Var}_R[T(d)] = \frac{1}{N^2}\sum_j \frac{\mu_j^2(d)}{p_j(1-p_j)}.$$

Studentization and normal limits

- **Conditions under $H_0(\tilde{d})$.** Independent Bernoulli assignments with $\epsilon \le p_i \le 1-\epsilon$; bounded potential outcomes and $|\mathcal{N}_i(4\bar{d})|$.
- Fixed tested distances, symmetric metric, and nondegenerate sampling and redraw covariance limits.
- **Consistent variance estimation:**

$$\frac{\widehat{\text{Var}}[T(d)]}{\text{Var}[T(d)]} \xrightarrow{p} 1.$$

- **Normal limits at a specified distance:**

$$T_s(d) = \frac{T(d)}{\sqrt{\widehat{\text{Var}}[T(d)]}} \rightsquigarrow \mathcal{N}(0,1),$$
$$F^*_{T_s(d)} \rightsquigarrow \mathcal{N}(0,1).$$

- The conditional randomization distribution $F^*_{T_s(d)}$ converges almost surely. Local-dependence approximation: Chen & Shao (2004). Re-studentize each redraw (Wu & Ding, 2021).

Joint testing with the Wald statistic

- Retain an intercept and distance-$0, \ldots, \tilde{d}$ controls. Test only the slopes at $\tilde{d}+1, \ldots, \bar{d}$:

$$T_{\text{Wald}} = \widehat{\boldsymbol{\beta}}'_{-\tilde{d}} \left(\widehat{\text{Var}}[\widehat{\boldsymbol{\beta}}_{-\tilde{d}}]\right)^{-1} \widehat{\boldsymbol{\beta}}_{-\tilde{d}}.$$

$$T_{\text{Wald}} \rightsquigarrow \chi^2_{\bar{d}-\tilde{d}}, \qquad F^*_{T_{\text{Wald}}} \rightsquigarrow \chi^2_{\bar{d}-\tilde{d}}.$$

- The conditional randomization distribution $F^*_{T_{\text{Wald}}}$ converges almost surely.

Asymptotic validity

Under $H_0(\tilde{d})$ and the stated conditions, both HT and Wald randomization tests satisfy, as $N, B \to \infty$,

$$\mathbb{P}(\widehat{p} \le \alpha) \longrightarrow \alpha, \qquad 0 < \alpha < 1.$$

Gaussian-prepivoted maximum

- A complementary joint test uses the maximum absolute studentized coefficient:

$$M_N^{\max} = \max_{1\le r\le \bar{d}-\tilde{d}} \frac{|\sqrt{N}\widehat{\beta}(\tilde{d}+r)|}{\sqrt{\widehat{V}_{N,rr}}}.$$

- Let $\mathsf{G}_N^{\max}$ be its fitted Gaussian percentile, with $\widehat{\mathbf{V}}_N = N\widehat{\text{Var}}[\widehat{\boldsymbol{\beta}}_{-\tilde{d}}]$; refit each redraw (Cohen & Fogarty, 2022).

$$p_{N,\max,\text{pre}} = \mathbb{P}^*(\mathsf{G}_N^{\max,*} \ge \mathsf{G}_N^{\max} \mid \boldsymbol{Z}, \boldsymbol{Y}).$$

- Under the corresponding Gaussian limits and consistent covariances, $p_{N,\max,\text{pre}} \rightsquigarrow \text{Unif}(0,1)$, even when the sampling and redraw covariance limits differ.

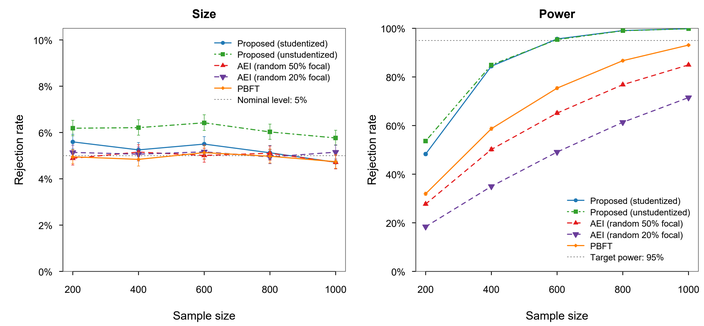
Simulation: comparison with focal-unit tests

20,000 replications | 999 redraws | 5% level

Cycle networks; $Z_i \overset{\text{ind}}{\sim} \text{Bern}(1/2)$; range R:

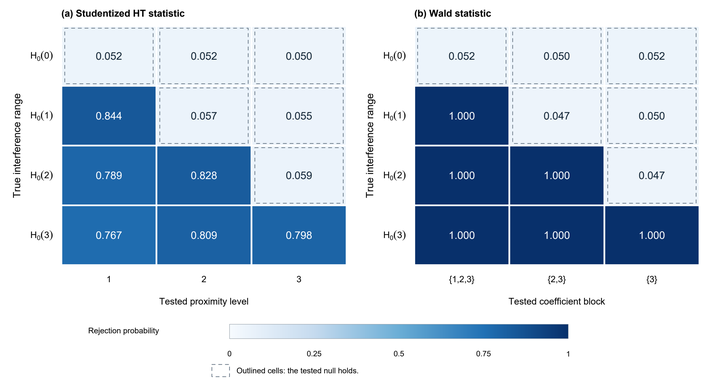
$$Y_i(z) = 5 + \varepsilon_i + (8+\eta_i)z_i + 2.85\sum_{h=1}^{R}(z_{i-h} + z_{i+h} - 1),$$

with $\varepsilon_i, \eta_i \overset{\text{iid}}{\sim} \text{Unif}[-\sqrt{3}, \sqrt{3}]$.



All methods test $H_0(0)$ at distance 1. AEI uses random 50%/20% focal sets. For PBFT, independent preliminary simulations select one of 20 prespecified focal sets, which is then held fixed for evaluation. Studentized HT size is 4.715–5.595%; mild over-rejection remains.

Simulation: testing multiple distances



$N = 400$; outlined cells satisfy the null. Scalar power: 76.655–84.430%; Wald: 100% in 20,000 repetitions. Three scalar null cells have size distortions. These simulations evaluate HT and Wald.

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weihan_zhang2001@berkeley.edu